



# CSCI 232

## 232 Algorithms & Data Structures

### LECTURE: 6

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# OUTLINE:

⚡ Structs

⚡ typedef

STRUCTURES

# STRUCTURES

Syntax:

```
struct symbolic_name1
{
    <statement1>
}<symbolic_name2, ...>;
```

- ✂ Structures are user-defined data types that group together variables of different data types.
- ✂ A structure is a collection of one or more variables, possibly of different types, grouped together under single `symbolic_name` for convenient handling.
- ✂ Everything that is in angle brackets `<>` is optional.

## TYPEDDEF & STRUCT 1/1

- ✖ Normally, when we define a struct, we can also declare global variables or pointers immediately after the closing brace:

```
struct Point
{
    int x, y;
} p1, *ptr;
```



- ✖ The variables `p1` and `ptr` are global objects of this structure type.

- ✖ We can also **omit the structure tag name** to make it **anonymous**, preventing multiple instances from being created:

```
struct
{
    int id;
    float value;
} config;
```



- ✖ Here, `config` is the **only instance** of this unnamed structure type.

## TYPEDEF & STRUCT 2/2

✂ Alternatively, we can use `typedef` to create an **alias** for a structure type:

```
typedef struct
{
    int x, y;
} Point;
int main()
{
    //...
    Point symbolic_name;
}
```



- ✂ Now we can declare variables as `Point p1;` — no need for the keyword `struct`.
- ✂ The name after the definition (`Point`) is a **type alias**, not a **variable**.
- ✂ **This alias does not prevent creating multiple instances** — it simply simplifies the syntax.

# TYPEDEF & STRUCT - SUMMARY

| Form                                       | Description                       | Example                                      |
|--------------------------------------------|-----------------------------------|----------------------------------------------|
| <code>struct Tag { ... } var;</code>       | Defines type + creates variable   | <code>struct Point {int x; int y;} p;</code> |
| <code>struct { ... } var;</code>           | Anonymous struct, single instance | <code>struct {int x;} s;</code>              |
| <code>typedef struct { ... } Alias;</code> | Creates a reusable type alias     | <code>typedef struct {int x;} Point;</code>  |

# AN EXAMPLE

```
#include <stdio.h>
struct MyStruct
{
    int value;
} *p, s; /* like struct MyStruct * sPointer; struct MyStruct myStructure; */

struct MyStruct * sPointer; /* global pointer */
struct MyStruct myStructure; /* global variable */

struct MyStruct function( struct MyStruct temp )
{
    temp.value += 5;
    return temp;
}
```



```
int main(int argc, char *argv[])
{
    struct MyStruct localStruct; /* local variable */
    localStruct.value = 1;
    struct MyStruct * localStructPointer; /* local pointer */
    localStructPointer = &localStruct;
    printf( "%d\n", localStruct.value ); // 1
    localStructPointer->value = 2;
    printf( "%d\n", localStructPointer->value ); // 2
    printf( "%d\n", (*localStructPointer).value ); /*Equivalent to the previous line*/ // 2
    p = &s;
    (*p).value = 2;
    printf( "%d\n", s.value ); // 2
    printf( "%d\n", p->value ); // 2

    printf( "%d\n", myStructure.value ); // 0
    myStructure = function( *localStructPointer );
    printf( "%d\n", myStructure.value ); // 7
    return 0;
}
```



## SUMMARY 1/3

- ✖ To access members of a structure, we use the dot (.) operator.
- ✖ When accessing a member through a pointer, we must use the dereferencing operator (\*) followed by the member access operator (.), enclosed in parentheses:  
`(*symbolic_name).field`.
- ✖ Alternatively, we can use the arrow operator `->`, which is equivalent to `symbolic_name->field`.

## SUMMARY 2/3

- ✂ Structures are user-defined data types that group together variables of different data types. They provide a way to create custom data types tailored to specific needs.
- ✂ **Components:** Structures consist of members (or fields), which can be of different data types.
- ✂ **Member access:**
  - ✂ **Direct:** To access members of a structure directly, use the dot operator (`.`).
  - ✂ **Indirect (through a pointer):** To access members through a pointer, use either:
    - ✂ the dereferencing operator (`*`) followed by the member access operator (`.`), enclosed in parentheses: `(*pointer).member`,
    - ✂ the arrow operator (`->`): `pointer->member`.

## SUMMARY 2/3

- ✂ **Pointers to structures:** You can create pointers to structures, similar to other data types. This allows dynamic memory allocation for structures and passing them as function arguments.
- ✂ **Global vs. local variables:** Structures can be declared as global variables (accessible from anywhere in the program) or local variables (accessible only within a specific block of code). You can also create pointers to structures for more flexible memory management.

THANK

*You*